



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHRI GURU SAI AND COMPANY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

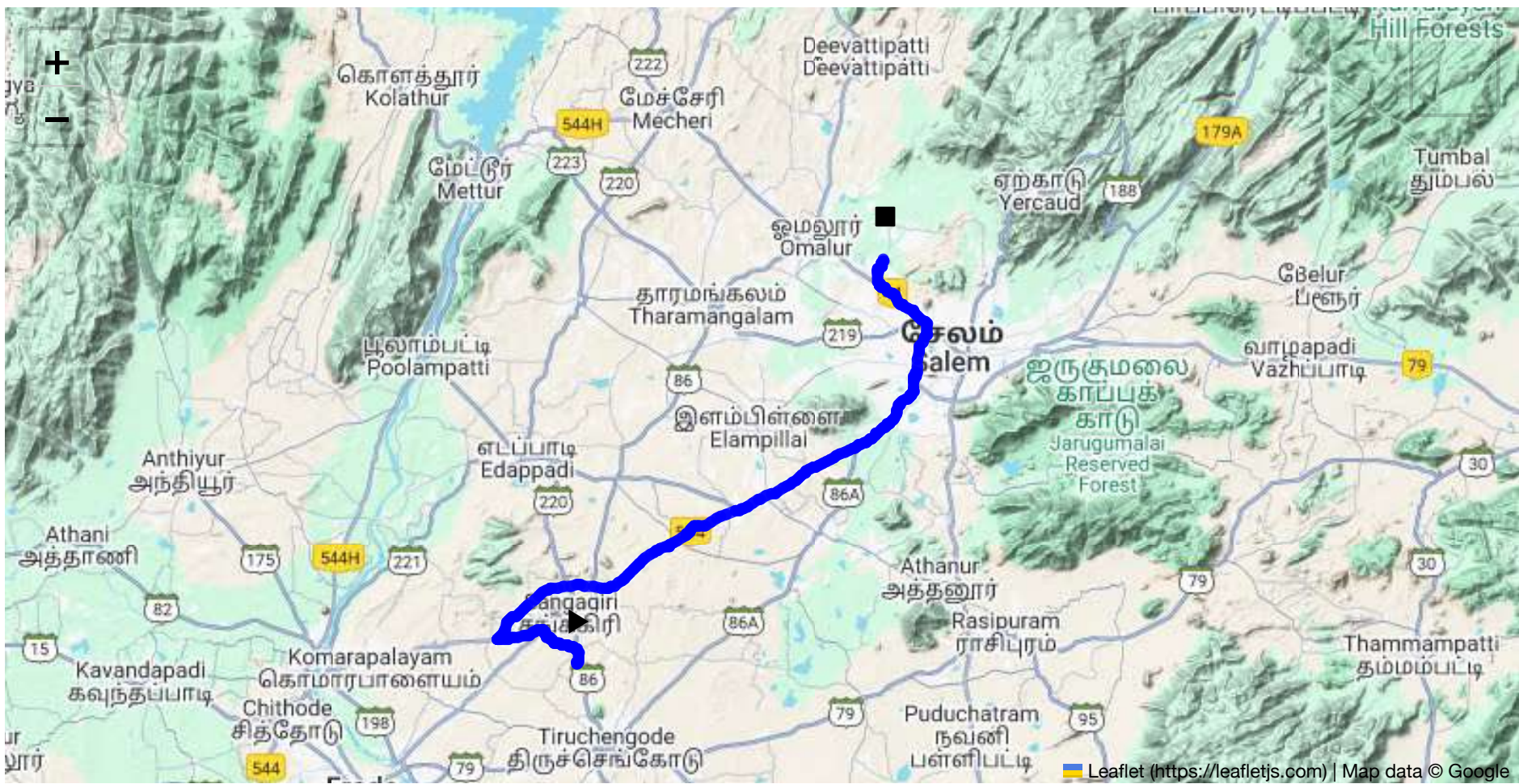
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 61.37 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours

Start: (11.4381, 77.8734)
End: (11.72577, 78.09626)



Welcome to the Journey Risk Management Study

- 1. Overview of the Route Map:** The route from CVQF+23W, Sangagiri, to P3GW+8F3, Karuppur, travels through key checkpoints like FR48+7XQ in Padaiveedu. It spans approximately 61.37 kilometers, primarily following the Bhavani Main Road, which is a significant connector in this region.
- 2. Typical Weather Conditions:** Tamil Nadu generally experiences a tropical climate with hot summers and heavy monsoon rains. During the monsoon season (June to September), roads can become slippery and waterlogged, increasing the risk of hydroplaning and reducing visibility. Heatwaves in summer may also affect vehicle performance.
- 3. Traffic Patterns Analysis:** The Bhavani Main Road is a busy corridor, particularly during the early morning (7-10 AM) and late afternoon (4-7 PM) due to commuting patterns. Traffic is notably heavier near major intersections and town areas. Congestion is common near Padaiveedu during peak hours.
- 4. Road Quality and Infrastructure:** Roads on this route are generally maintained but vary in quality. Expect some potholes and uneven surfaces, especially after monsoon rains, which can further deteriorate road conditions. Urban areas may have better-maintained roads compared to rural stretches.
- 5. Alternative Routes for Emergencies:** In case of roadblocks or accidents, the NH544 (Salem-Kochi Highway) can act as an alternate route. Detours via local roads, though longer, might be necessary depending on the nature of the emergency.
- 6. Local Regulations on Hazardous Material Transport:** Tamil Nadu has stringent rules governing the transportation of hazardous materials, including specific permits and clear labeling. Drivers should be acquainted with local requirements and ensure compliance with safety protocols such as spill kits and proper training.
- 7. Historical Incidents Involving Heavy Vehicles:** Accidents involving heavy vehicles have been reported, especially around congested areas due to sudden braking or poor visibility conditions. The region has seen incidents due to vehicle overloading and mechanical failures.

8. Environmental Considerations: The route passes through agricultural zones, where potential chemical spills could impact soil and water quality. Care should be taken to avoid contamination, and handlers should be prepared for emergency cleanup.

9. Communication Coverage: The route is generally well-covered with cellular networks, but rural expanses between population centers may have coverage issues. Ensure communication devices are fully charged and consider using supplementary satellite communication if needed.

10. Emergency Response Times: Emergency services are typically accessible within 30-45 minutes in urban centers, but response times can increase significantly, up to an hour or more, in rural areas due to distance and less accessible terrain.

12. Overall Summary of Risk Assessment: Transporting hazardous materials on this route is moderately risky due to weather variability, traffic congestion, and road quality issues. Conscientious planning, adherence to regulations, and readiness for emergencies are essential to mitigating these risks. Training drivers in navigation, communication, and spill control can further enhance safety. Implementing a real-time tracking system with emergency response contacts is advisable.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45358, 77.85781	30 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46053, 77.85034	30 KM/Hr
7	Turn	Medium	11.46307, 77.84941	30 KM/Hr
8	Turn	Medium	11.46316, 77.84921	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
11	Turn	Medium	11.64183, 78.11810	30 KM/Hr
12	Turn	Medium	11.68920, 78.11799	30 KM/Hr
13	Turn	Medium	11.69386, 78.10968	30 KM/Hr
14	Turn	Medium	11.69392, 78.10964	30 KM/Hr
15	Turn	High	11.69400, 78.10963	15 KM/Hr
16	Turn	Medium	11.70817, 78.09316	30 KM/Hr
17	Turn	High	11.70831, 78.09326	15 KM/Hr
18	Turn	Medium	11.70850, 78.09305	30 KM/Hr
19	Turn	Medium	11.71612, 78.09288	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	High	11.71689, 78.09240	15 KM/Hr
21	Turn	High	11.71764, 78.09254	15 KM/Hr
22	Turn	Medium	11.71813, 78.09187	30 KM/Hr
23	Turn	High	11.71825, 78.09182	15 KM/Hr
24	Turn	Medium	11.72096, 78.09476	30 KM/Hr
25	Turn	High	11.72211, 78.09521	15 KM/Hr
26	Turn	High	11.72251, 78.09496	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
15	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
16	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
20	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
21	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
23	hospital	Rajaam Hospital	11.7212623, 78.0951467	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
19	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



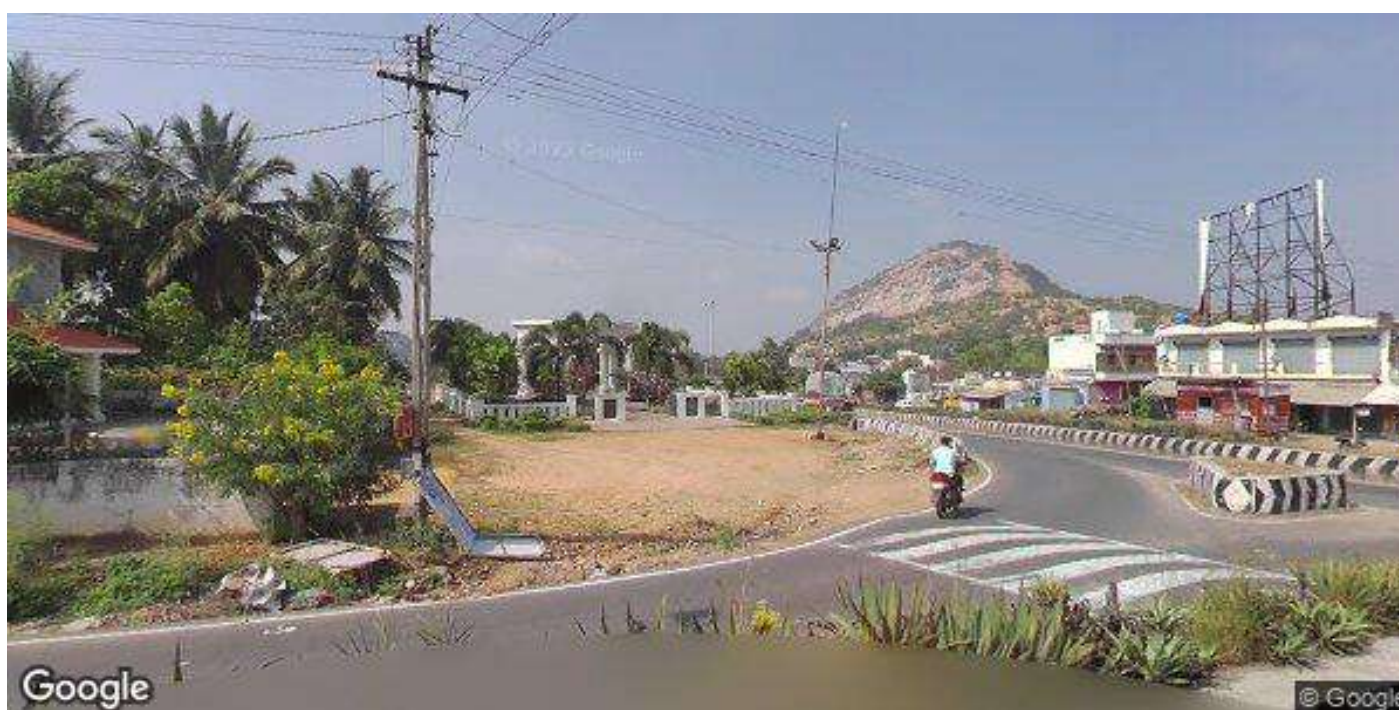
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46053, 77.85034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68920, 78.11799



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69386, 78.10968



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69392, 78.10964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69400, 78.10963



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70817, 78.09316



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70831, 78.09326



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70850, 78.09305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71612, 78.09288



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.71689, 78.09240



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.71764, 78.09254



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71813, 78.09187



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.71825, 78.09182



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72096, 78.09476



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72211, 78.09521



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72251, 78.09496

